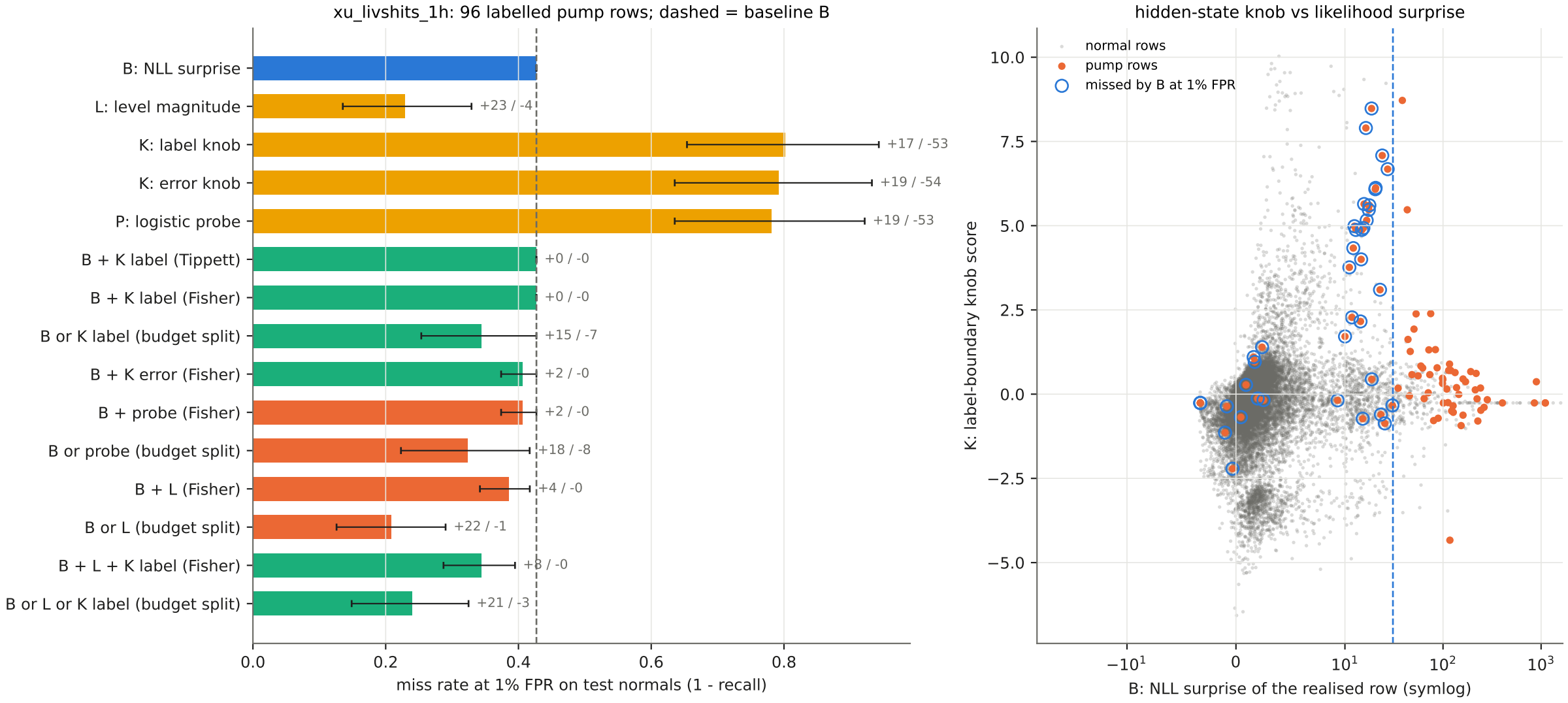

xu_livshits_1h: 96 labelled pump rows; dashed = baseline B
B: NLL surprise
L: level magnitude
+23 / -4
K: label knob
+17 / -53
K: error knob
+19 / -54
P: logistic probe
+19 / -53
B + K label (Tippett)
+0 / -0
B + K label (Fisher)
+0 / -0
B or K label (budget split)
+15 / -7
B + K error (Fisher)
+2 / -0
B + probe (Fisher)
+2 / -0
B or probe (budget split)
+18 / -8
B + L (Fisher)
+4 / -0
B or L (budget split)
+22 / -1
B + L + K label (Fisher)
+8 / -0
B or L or K label (budget split)
+21 / -3
0.0
0.2
0.4
0.6
0.8
miss rate at 1% FPR on test normals (1 - recall)
hidden-state knob vs likelihood surprise
normal rows
pump rows
missed by B at 1% FPR
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
K: label-boundary knob score
−10^1
0
10^1
10^2
10^3
B: NLL surprise of the realised row (symlog)